AlgoRhythms
(NariY_11)

SentimentQ

SDG 3: Good Health and Well being

Team Members: Tanisha Das, Swetha R, Swapnali Singh

College: Christ University, Bengaluru

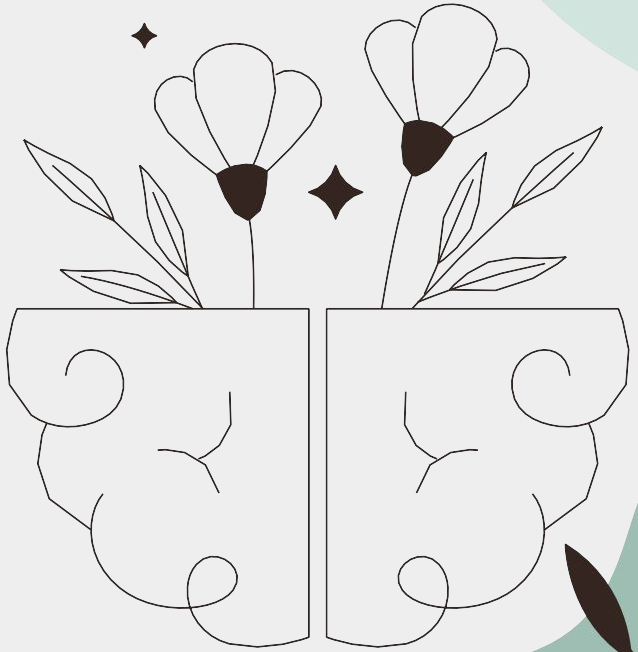


Table of contents

SL No.	Description
1.	Problem Statement and Motivation
2.	Project Objectives
3.	System Architecture
4.	Quantum Search with Grover's Algorithm
5.	Sentiment analysis using VADER model
6.	Technical Stack
7.	Results

Problem Statement and Motivation

Social media platforms generate large volumes of text containing discussions about health and well-being. Extracting relevant information from these collections requires efficient search and subsequent text analysis.

Challenges

- Searching large tweet collections can require substantial computational effort.
- Health-related content must be identified before meaningful sentiment analysis can be performed.
- Search performance needs to be evaluated against an appropriate classical baseline.

Research question

Can Grover's quantum search algorithm be integrated into a social media analysis pipeline, and how does its measured performance compare with classical search?

This is stronger than saying that manual filtering is difficult because it identifies the computational question your project actually explores.

Project Objectives

- Apply Grover's algorithm to search for relevant records in a large tweet dataset.
- Analyse the sentiment of retrieved tweets using classical Natural Language Processing (NLP).
- Compare search performance against a classical baseline.
- Visualise retrieved tweets and their sentiment distributions through an interactive dashboard.



Quantum Computing

Prioritize tweets containing health-related keywords using Grover’s algorithm



NLP

Perform sentiment analysis on the filtered tweets to classify them as ‘**positive**’, ‘**negative**’.



Visualization Dashboard

Present insights in a user-friendly interface using charts and tables.

System Architecture



Quantum Search with Grover's Algorithm

How Grover's Algorithm Works

- State Preparation: Represent the candidate search space as quantum states.
- Oracle: Mark states that satisfy the search condition.
- Amplitude Amplification: Increase the probability of measuring a marked state.
- Measurement: Obtain a candidate search result.

Theoretical Query Complexity

- Classical exhaustive search: $O(N)$
- Grover's algorithm: $O(\sqrt{N})$

Here, (N) represents the number of candidates in an unstructured search space.



Sentiment Analysis Using NLP

Extracting Sentiment from Retrieved Tweets

- Process the text of relevant tweets using the implemented NLP pipeline.
- Assign sentiment labels based on the textual content.
- Summarise sentiment distributions across the retrieved records.
- Visualise the results using charts and tables.

Output: Retrieved tweet records, sentiment classifications, and sentiment-distribution visualisations.

Evaluating Search Performance

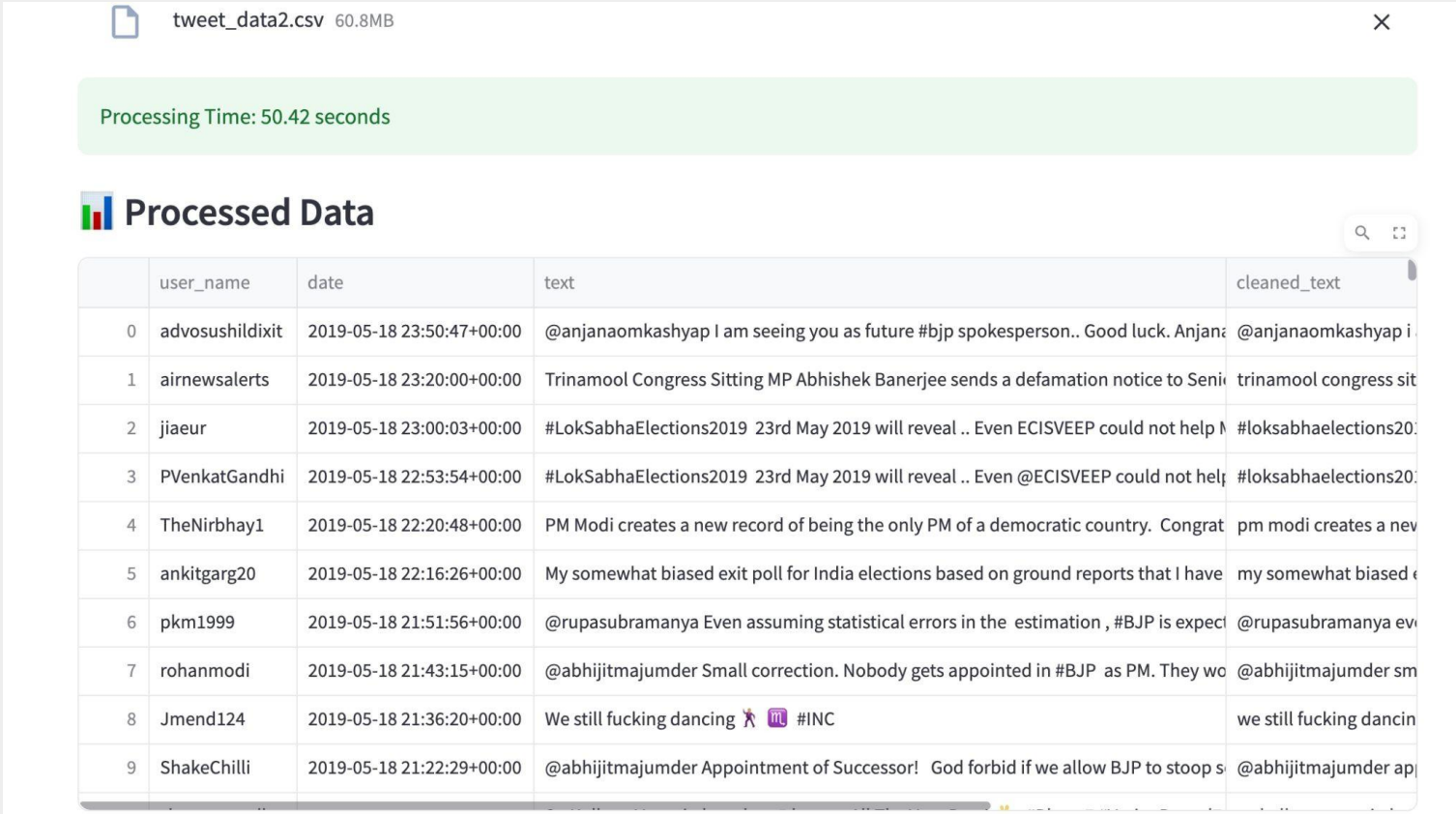
- Dataset: Over 3 million tweet records.
- Classical baseline: Classical search implementation.
- Quantum approach: Grover's algorithm implemented using Qiskit.
- Evaluation metric: Elapsed time for the measured operation.

Technical Stack

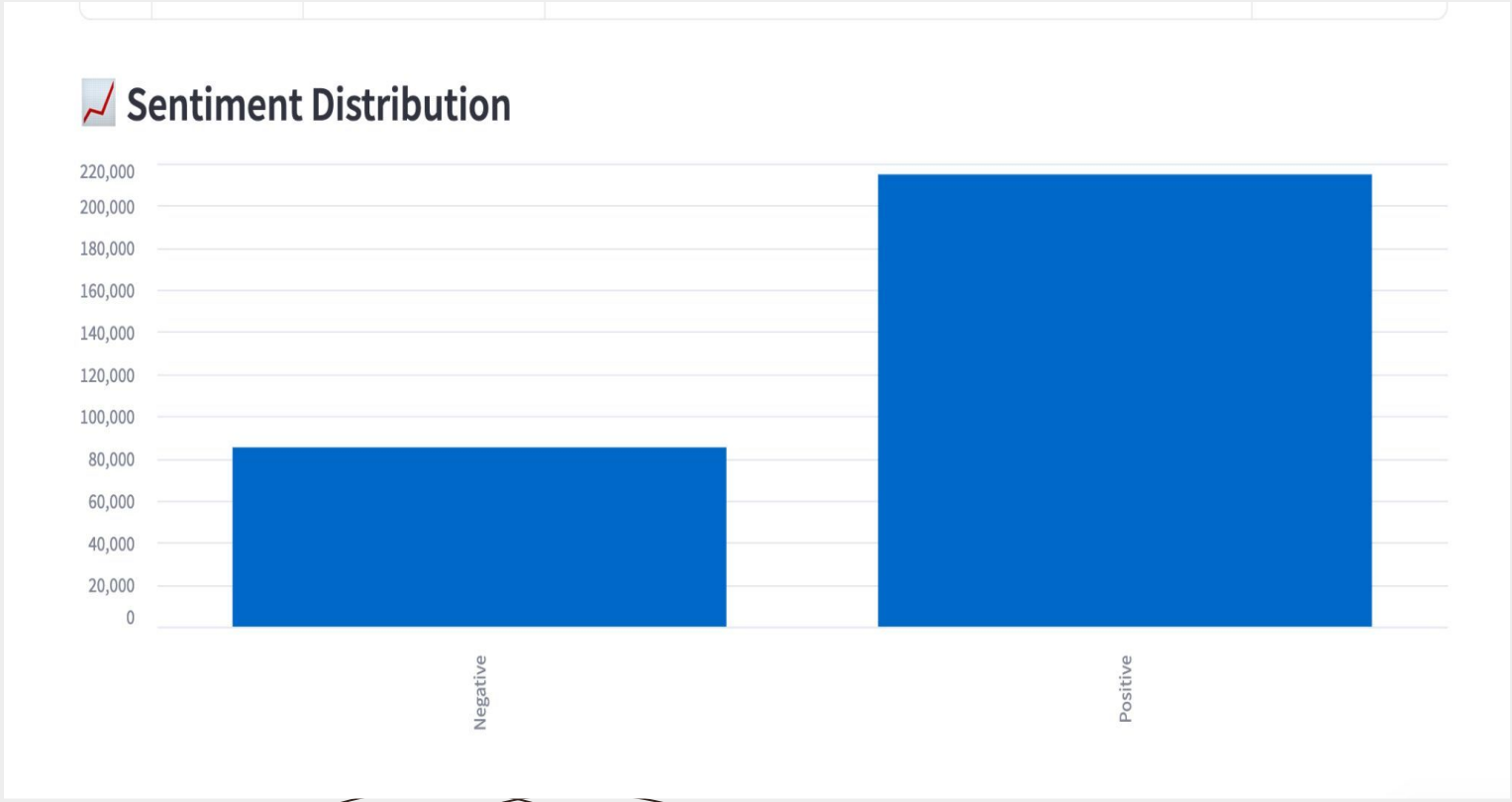
Component	Library Used	Justification
Programming	Python	Core programming and application logic.
Quantum Search	Qiskit	Quantum circuit implementation and Grover's algorithm.
Sentiment Analysis	VADER from NLTK	Tweet sentiment analysis and text processing.
Frontend	Streamlit	Interactive dashboard.

Results

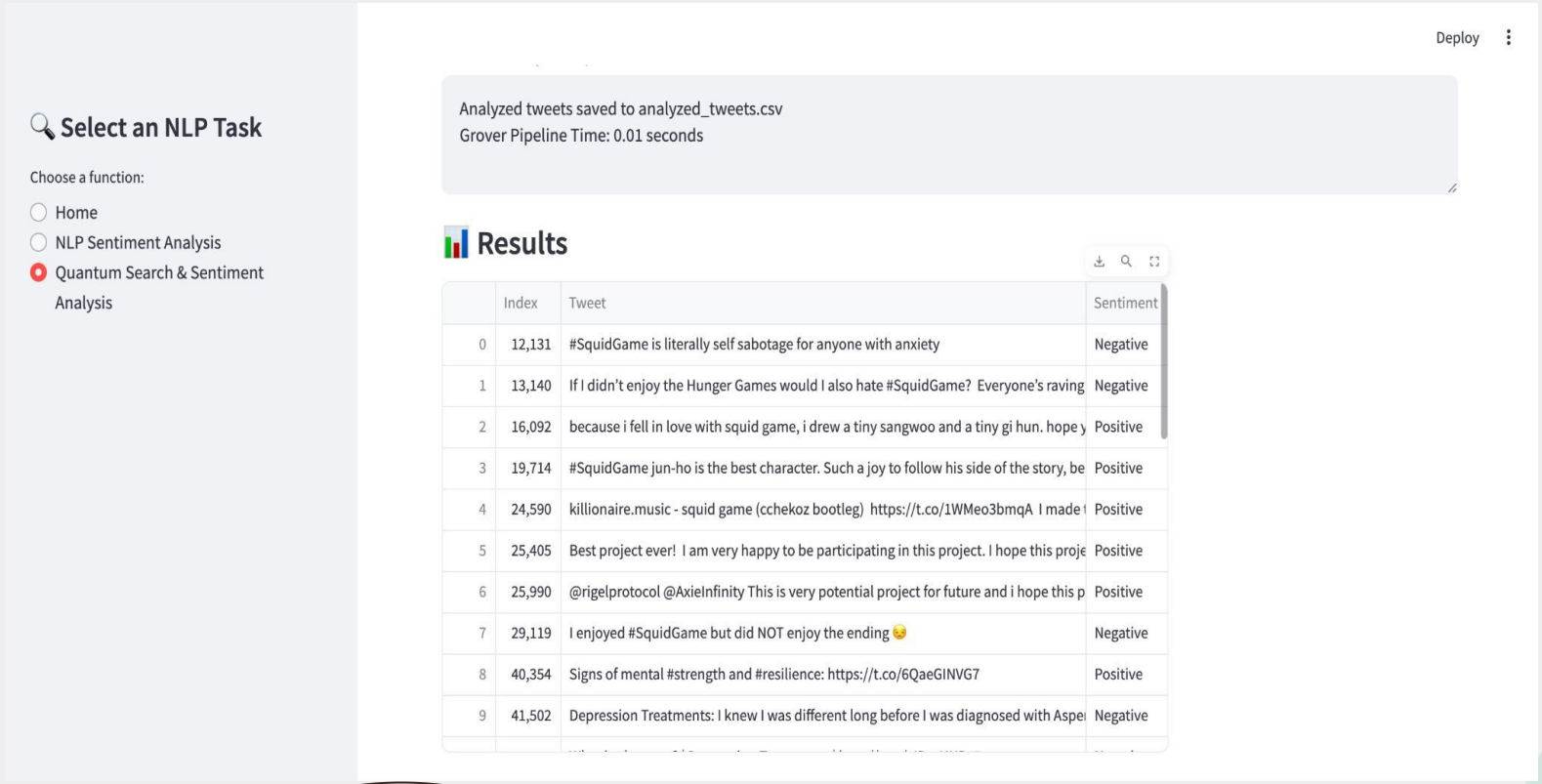
Linear Search

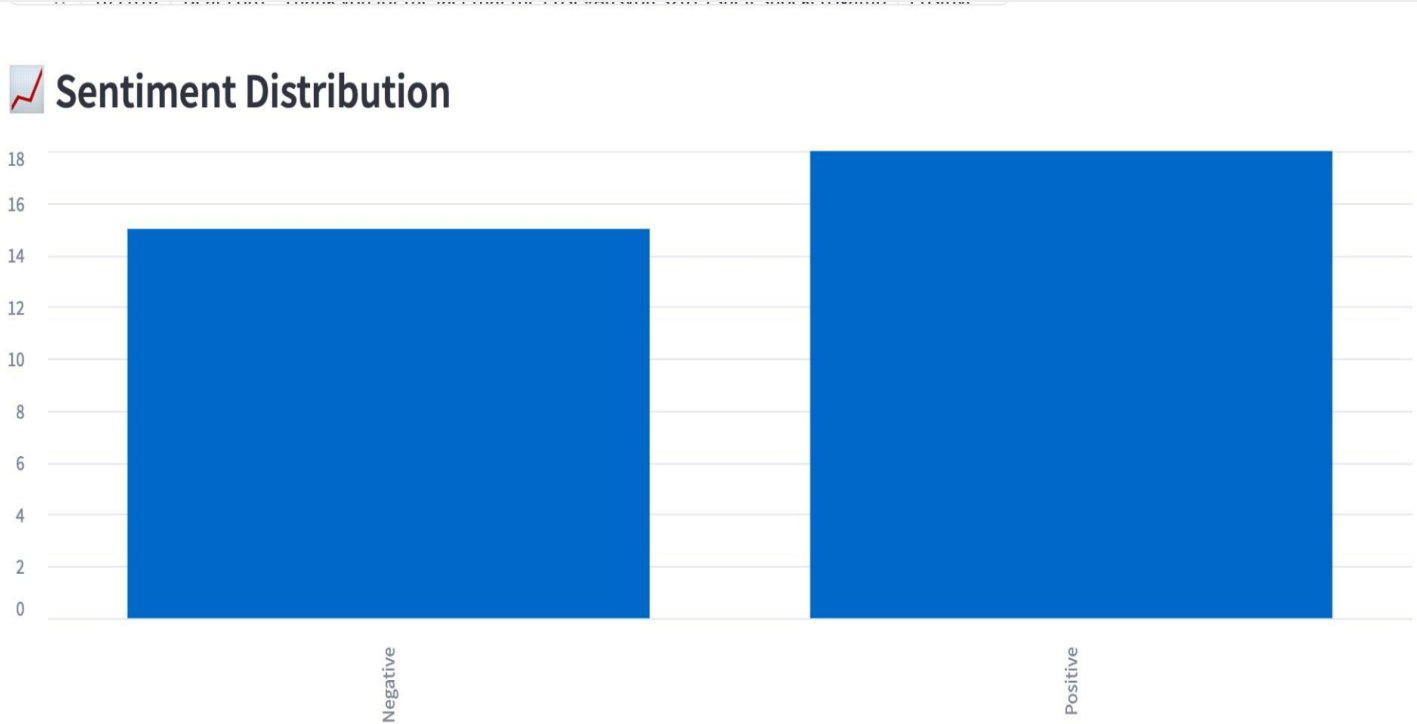


	user_name	date	text	cleaned_text
0	advosushildixit	2019-05-18 23:50:47+00:00	@anjanaomkashyap I am seeing you as future #bjp spokesperson.. Good luck. Anjana	@anjanaomkashyap i
1	airnewsalerts	2019-05-18 23:20:00+00:00	Trinamool Congress Sitting MP Abhishek Banerjee sends a defamation notice to Senio	trinamool congress sit
2	jiaeur	2019-05-18 23:00:03+00:00	#LokSabhaElections2019 23rd May 2019 will reveal .. Even ECISVEEP could not help M	#loksabhaelections20
3	PVenkatGandhi	2019-05-18 22:53:54+00:00	#LokSabhaElections2019 23rd May 2019 will reveal .. Even @ECISVEEP could not help	#loksabhaelections20
4	TheNirbhay1	2019-05-18 22:20:48+00:00	PM Modi creates a new record of being the only PM of a democratic country. Congrat	pm modi creates a nev
5	ankitgarg20	2019-05-18 22:16:26+00:00	My somewhat biased exit poll for India elections based on ground reports that I have	my somewhat biased e
6	pkm1999	2019-05-18 21:51:56+00:00	@rupasubramanya Even assuming statistical errors in the estimation , #BJP is expect	@rupasubramanya ev
7	rohanmodi	2019-05-18 21:43:15+00:00	@abhijitmajumder Small correction. Nobody gets appointed in #BJP as PM. They wo	@abhijitmajumder sm
8	Jmend124	2019-05-18 21:36:20+00:00	We still fucking dancing #INC	we still fucking dancin
9	ShakeChilli	2019-05-18 21:22:29+00:00	@abhijitmajumder Appointment of Successor! God forbid if we allow BJP to stoop s	@abhijitmajumder ap


Sentiment Distribution
220,000
200,000
180,000
160,000
140,000
120,000
100,000
80,000
60,000
40,000
20,000
0
Negative
Positive

Grover Search




Sentiment Distribution
18
16
14
12
10
8
6
4
2
0
Negative
Positive

Thank You